

Half-plane intersection

In this article we will discuss the problem of computing the intersection of a set of half-planes. Such an intersection can be conveniently represented as a convex region/polygon, where every point inside of it is also inside all of the half-planes, and it is this polygon that we're trying to find or construct. We give some initial intuition for the problem, describe a $O(N \log N)$ approach known as the Sort-and-Incremental algorithm and give some sample applications of this technique.

It is strongly recommended for the reader to be familiar with basic geometrical primitives and operations (points, vectors, intersection of lines). Additionally, knowledge about [Convex Hulls](#) or the [Convex Hull Trick](#) may help to better understand the concepts in this article, but they are not a prerequisite by any means.

Initial clarifications and definitions

For the entire article, we will make some assumptions (unless specified otherwise):

1. We define N to be the quantity of half-planes in the given set.
2. We will represent lines and half-planes by one point and one vector (any point that lies on the given line, and the direction vector of the line). In the case of half-planes, we assume that every half-plane allows the region to the left side of its direction vector. Additionally, we define the angle of a half-plane to be the polar angle of its direction vector. See image below for example.
3. We will assume that the resulting intersection is always either bounded or empty. If we need to handle the unbounded case, we can simply add 4 half-planes that define a large-enough bounding box.
4. We will assume, for simplicity, that there are no parallel half-planes in the given set. Towards the end of the article we will discuss how to deal with such cases.